T-1

Solve all the questions.

1. Materials for a Constructions site are supplied by two sources A and B. Upon arrival of all materials from both sources, the Construction is started after the arrival of workers. If the probability that meterials would be supplied by A on time is 0.8 and by B is 0.75, and if the probability that the workers arrive on time for work is 0.95, find the probability that the job is started on time.

2. Find the following Conditional probabilities:

a. On a single roll of a die, the probability of a 3 given that the value is an odd integer (draw the sample space for the conditional event).

b. The probability that the outcome of a single draw from a card deck is a red ace given that it is not a face card.

3. A construction site receives fill material from three sources, with sources A, B and C providing 15, 25 and 60% of total, respectively. On the average, fill material from sources A, B and C do not have the proper moisture characteristics 2, 6 and 3% on the time, respectively. If one dump truck is sampled at random and the fill material does not have the proper moisture what is the probability that it came
(a) from Source A
(b) from Source B, and
(c) from Source C ?